

GUIDED LESSON • LEARN BY EXPERIMENTING

Global & Relative Phase

A from-scratch walkthrough for total beginners: what phase really means, why global phase is invisible, why relative phase matters, and how applying H twice brings a qubit back to where it started.

00 Before You Begin

Four ideas you need first

1. A qubit has a state

A classical bit is either **0** or **1**. A qubit can be in a combination of both. We write the basic states as $|0\rangle$ and $|1\rangle$.

2. Superposition uses amplitudes

A qubit can be written as $\alpha|0\rangle + \beta|1\rangle$. The numbers α and β are called **amplitudes**.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

α and β describe the qubit's quantum state

3. Amplitudes contain more information than probabilities

When we measure a qubit, we see probabilities. But before measurement, amplitudes also contain **phase information**.

The important distinction

Probability tells us **how likely** an outcome is. Phase helps determine **how quantum states interfere**.

4. Phase does not mean "probability"

This is one of the most important beginner ideas. Two quantum states can have exactly the same measurement probabilities but behave differently when a quantum gate is applied.

01

What Is Phase?

Think of phase as a hidden quantum "angle"

1. Start with a simple complex number

In quantum computing, amplitudes can be **complex numbers**. A complex number can be thought of as having a magnitude and an angle.

$$z = r e^{i\theta}$$

r = magnitude | θ = phase

2. A phase can change the amplitude without changing its size

Consider the two amplitudes:

$$+1 \quad \text{and} \quad -1$$

Same magnitude, different phase

Both have magnitude 1, so squaring the magnitude gives the same probability:

$$|+1|^2 = 1 \quad |-1|^2 = 1$$

Beginner takeaway

Phase can be invisible when you measure immediately, but it can become visible after later gates cause interference.

3. The simplest phase change: a minus sign

Compare these two states:

$$|\psi_1\rangle = |0\rangle + |1\rangle$$

$$|\psi_2\rangle = |0\rangle - |1\rangle$$

Their probabilities are the same because the sign disappears when we calculate the squared magnitude. But the relative phase between $|0\rangle$ and $|1\rangle$ is different.

Global Phase

When the whole state gets the same phase

1. What does "global" mean?

Global means the phase is applied to the **entire quantum state**.

Suppose we have:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

If we multiply the entire state by the same phase factor $e^{i\varphi}$, we get:

$$|\psi'\rangle = e^{i\varphi} (\alpha|0\rangle + \beta|1\rangle)$$

2. Example: multiply everything by -1

Take:

$$|\psi\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

Multiply the **whole state** by -1 :

$$-|\psi\rangle = (-|0\rangle - |1\rangle) / \sqrt{2}$$

Every amplitude received the same phase change. This is a **global phase**.

3. Can we detect global phase?

Not by measuring that isolated quantum state. Global phase does not change ordinary measurement probabilities.

$$|e^{i\varphi} \alpha|^2 = |\alpha|^2$$

The phase factor has magnitude 1

Global phase → physically unobservable by itself

It changes the mathematical description of the state, but not the measurement probabilities of that isolated state.

03 Relative Phase

The phase difference between parts of a state

1. "Relative" means comparing two amplitudes

Suppose our qubit is:

$$|\psi\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

Now compare it with:

$$|\varphi\rangle = (|0\rangle - |1\rangle) / \sqrt{2}$$

The $|1\rangle$ part has a minus sign relative to the $|0\rangle$ part. This is a **relative phase of π radians** or **180°**.

STATE A

Same phase

$$(|0\rangle + |1\rangle) / \sqrt{2}$$

The two amplitudes point in the same phase direction.

STATE B

Relative phase π

$$(|0\rangle - |1\rangle) / \sqrt{2}$$

The two amplitudes differ by 180°.

2. Why is relative phase important?

Because quantum gates can make amplitudes **add** or **cancel**. This is called **interference**.

The key difference

Global phase = same phase applied everywhere.

Relative phase = phase difference between components.

Relative phase can affect future measurement results because it changes interference.

3. A very useful comparison

Concept	What happens?	Can it affect interference?
Global phase	The entire state gets the same phase.	Not by itself.
Relative phase	One component has a different phase from another.	Yes.

04 The $H \rightarrow H$ Experiment

Apply Hadamard twice and watch what happens

1. First: what is H?

H stands for the **Hadamard gate**. It is one of the most important gates in quantum computing.

Its main job for a beginner is to create and manipulate **superposition and relative phase**.

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Hadamard gate matrix

2. What does H do to $|0\rangle$?

Applying H to $|0\rangle$ gives an equal superposition:

$$H|0\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

Therefore:

$$P(0) = 1/2 \quad P(1) = 1/2$$

50% chance of 0 and 50% chance of 1

Circuit 1 • One H gate

$|0\rangle$



3. Now apply H one more time

The interesting experiment is:

Circuit 2 • H → H

$|0\rangle$

H

H

M

4. Step-by-step calculation

Start with:

$$|\psi_0\rangle = |0\rangle$$

First H:

$$H|0\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

Now apply the second H:

$$H [(|0\rangle + |1\rangle) / \sqrt{2}]$$

5. Apply H to each component

We know:

$$H|0\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

$$H|1\rangle = (|0\rangle - |1\rangle) / \sqrt{2}$$

Therefore:

$$H [(|0\rangle + |1\rangle) / \sqrt{2}] = 1/\sqrt{2} [H|0\rangle + H|1\rangle]$$

6. Substitute the two results

$$= 1/\sqrt{2} [(|0\rangle + |1\rangle)/\sqrt{2} + (|0\rangle - |1\rangle)/\sqrt{2}]$$

Combine them:

$$= 1/2 [|0\rangle + |1\rangle + |0\rangle - |1\rangle]$$

The $+|1\rangle$ and $-|1\rangle$ terms cancel:

$$= 1/2 [2|0\rangle]$$

$$= |0\rangle$$

$$H \rightarrow H \rightarrow |0\rangle$$

Two Hadamard gates undo each other. The qubit returns to its original state.

05

Why Did $H \rightarrow H$ Work?

The answer is interference

1. The first H creates two paths

After the first H, the qubit is:

$$(|0\rangle + |1\rangle) / \sqrt{2}$$

We can imagine two quantum "paths":

PATH A

Through $|0\rangle$

The second H sends $|0\rangle$ toward both $|0\rangle$ and $|1\rangle$.

PATH B

Through $|1\rangle$

The second H sends $|1\rangle$ toward $|0\rangle$ and $|1\rangle$ too, but with a minus sign.

RESULT

Interference

The $|1\rangle$ contributions cancel while the $|0\rangle$ contributions reinforce.

2. Constructive interference

The contributions toward $|0\rangle$ have the same sign, so they **add together**.

$$+1/2 + +1/2 = +1$$

Contributions reinforce each other

3. Destructive interference

The contributions toward $|1\rangle$ have opposite signs, so they **cancel**.

$$+1/2 + -1/2 = 0$$

Contributions cancel each other

This is the important connection

The relative minus sign created by the Hadamard transformation is what allows the second H to produce cancellation and reinforcement.

4. The complete picture

$$|0\rangle \rightarrow H \rightarrow (|0\rangle + |1\rangle)/\sqrt{2} \rightarrow H \rightarrow |0\rangle$$

5. The matrix shortcut

Because H is its own inverse:

$$H \times H = I$$

I = identity matrix

Therefore applying H twice is equivalent to doing nothing:

$$H^2 = I$$



Quick Revision

The five things to remember

- **Phase** describes the angular information contained in a quantum amplitude.
- **Global phase** multiplies the entire quantum state by the same phase factor and cannot be observed through ordinary measurement of that isolated state.
- **Relative phase** describes the phase difference between components such as $|0\rangle$ and $|1\rangle$.
- **Relative phase matters** because it can control constructive and destructive interference.
- **$H \rightarrow H = I$** , so applying the Hadamard gate twice returns a qubit to its original state.

$$H^2 = I$$

First H creates a superposition. Second H uses interference to reconstruct the original state.

One-line memory trick

Global phase doesn't matter by itself; relative phase controls interference; H followed by H cancels out.